

# Pathlavath Shiva Kumar

AI/ML | Data Analysis | Full-Stack Development

Hyderabad, India

✉ [pathlavathshivakumar978@gmail.com](mailto:pathlavathshivakumar978@gmail.com) [in LinkedIn](#) [GitHub](#) [Portfolio](#)

## Education

Indian Institute of Information Technology Vadodara

B.Tech in Computer Science and Engineering

Aug 2023 – May 2027

Expected graduation: May 2027

## Technical Skills

**Languages:** Python, JavaScript, Java, C, SQL

**AI/ML and Data:** PyTorch, OpenCV, scikit-learn, XGBoost, SHAP, Pandas, NumPy, Plotly

**Retrieval and Applications:** ChromaDB, Sentence Transformers, BM25, Streamlit, Groq API

**Web and Tools:** React, Node.js, Express, MongoDB, Socket.IO, Git, GitHub

**Core CS:** Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

## Experience

Xerlon — Software Development Engineer Intern

Jul 2026 – Aug 2026

- Contributed to code implementation, testing and delivery for assigned engineering projects during a completed software development internship.

Unified Mentor Pvt. Ltd. — Data Analyst Intern

May 2026 – Jul 2026

- Prepared customer data, engineered features and compared **five classifiers**; built a **Streamlit** churn-risk dashboard with **SHAP** explanations.
- Developed **Pandas/Plotly** profitability views for Nassau Candy to analyze product revenue, gross profit, margins and regional performance.

## Projects

PDF RAG Assistant

GitHub

*Python, Streamlit, ChromaDB, Groq*

- Built multi-PDF question answering with **PyMuPDF**, page metadata and **800-character chunks/100-character overlap**; integrated streamed **Groq** responses into Streamlit chat.
- Combined **BM25 and BGE/ChromaDB retrieval** (top 10 each), deduplicated candidates and cross-encoder reranked to **five context passages** with source-citation prompts.

Driver Drowsiness Detection

GitHub

*Python, PyTorch, torchvision, OpenCV*

- Developed a **Darknet-53/attention classifier** for drowsy/non-drowsy images; used **66,521 images** with resizing, normalization, flipping, rotation and color augmentation.
- Trained with **SGD and cross-entropy**, recording **89.03% validation accuracy** on an **80/20 random image split**; added confidence scores, webcam inference and **Grad-CAM**.

Bank Customer Churn Dashboard

GitHub | Live Demo

*scikit-learn, SHAP, Streamlit*

- Engineered features on **10,000 customer records**, compared **five classifiers** and built **Streamlit/SHAP** views for churn-risk estimates and what-if analysis.
- Recorded **ROC-AUC 0.869, F1 0.604 and accuracy 86.95%** with gradient boosting on the **2,000-record holdout** used for model comparison (stratified 80/20 split).

MERN Real-Time Chat Application

GitHub | Live Demo

*React, Node.js, MongoDB*

- Built **React/Express** registration, login and chat views with **Socket.IO** live messaging and **MongoDB** conversation persistence.
- Implemented **JWT**-protected HTTP routes and **bcrypt** password hashing for user authentication.

## Achievements

- LeetCode contest rating: 1500+**; solved **500+ DSA problems** across coding platforms.

## Certifications

Fundamentals of Deep Learning — NVIDIA

14 October 2025

Certificate of Competency

View verified credential